

Physics (OG)

Motion in a Straight Line

Lecture-01

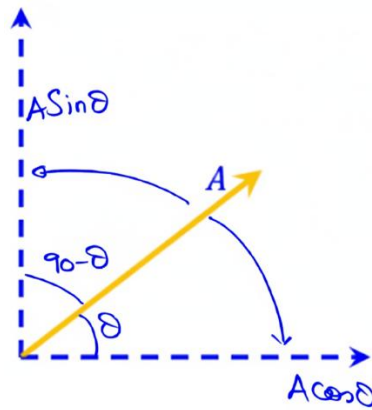
Concepts Covered in the Class:

- Rectangular Component
- Scalar Product (dot product)
- Dot Product is use to calculate Projection of vector
- Motion
- Observer and Object
- Position of B w.r.t. A ($r_{B/A}$)
- Displacement (S) and Distance travelled (I)
- Important points

Rectangular Component

- Any vector can be broken down into perpendicular components, typically along x and y axes.
- These components are calculated using trigonometry: $A \sin\theta$ and $A \cos\theta$, where θ is the angle with the x -axis.
- The resultant of two perpendicular vectors P and Q is given by:
$$Net = \sqrt{P^2 + Q^2}.$$



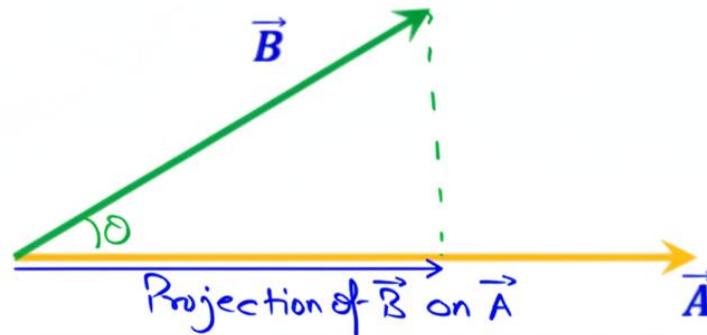


Scalar Product (dot product)

- The scalar product (dot product) of two vectors A and B is defined as $A \cdot B = AB \cos \theta$.
- If two vectors are perpendicular ($\theta = 90^\circ$), their dot product is zero ($A \cdot B = 0$).
- The dot product can be used to calculate the angle between two vectors: $\cos \theta = (A \cdot B) / AB$.

Dot Product is use to calculate Projection of vector

- The projection of vector B onto vector A is given by $(B \cos \theta) \hat{A}$ or $(A \cdot B) / A^2$.
- The projection represents the component of one vector that lies along the direction of another vector.

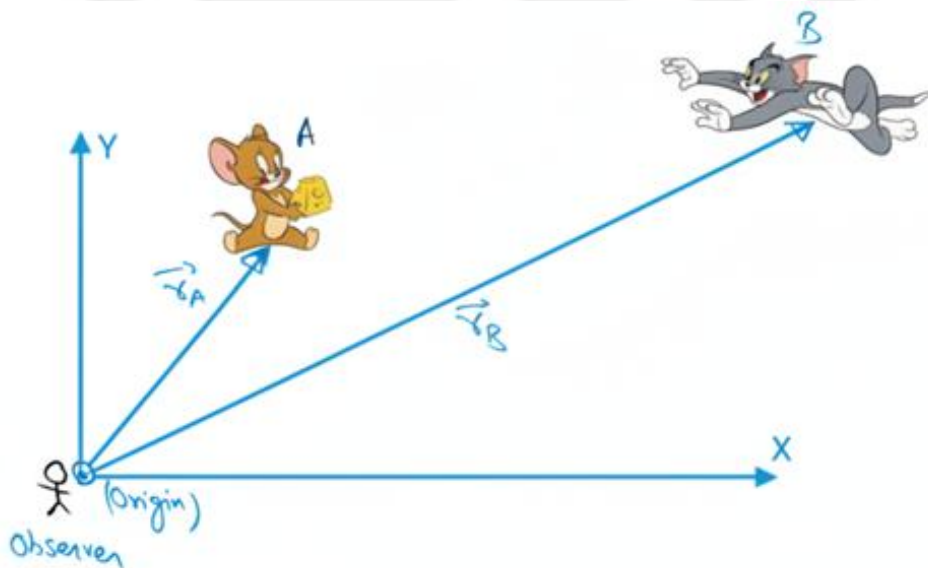


$$\begin{aligned} \text{Projection of } \vec{B} \text{ on } \vec{A} &= (B \cos \theta) \hat{A} \\ &= (\vec{A} \cdot \vec{B}) \frac{\vec{A}}{A^2} \end{aligned}$$

$$\text{Projection of } \vec{A} \text{ on } \vec{B} = (\vec{A} \cdot \vec{B}) \frac{\vec{B}}{B^2}$$

Motion

- Motion is relative and always depends on the chosen observer and their frame of reference (origin).
- To define motion, an observer and an origin point are essential.
- The position vector of an object from the origin describes its location.



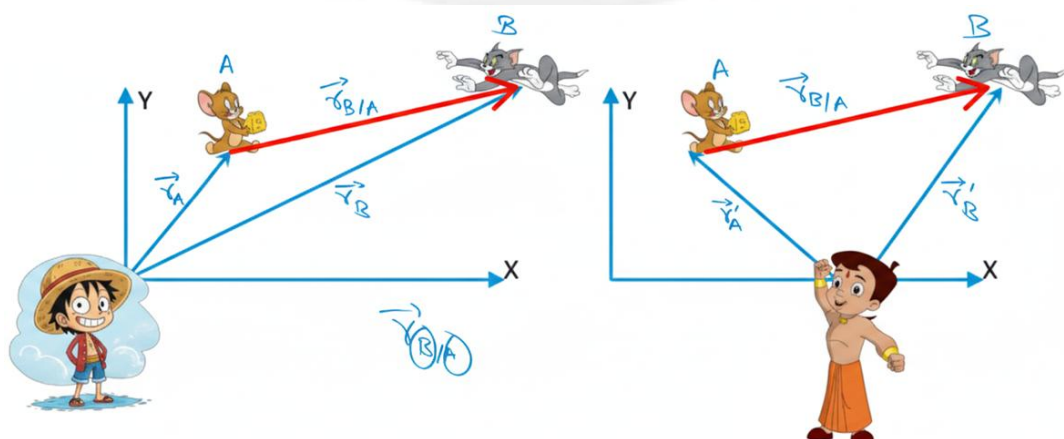
Observer and Object

- An object can be at rest relative to one observer but in motion relative to another.
- For example, a passenger in a car is at rest relative to the car's driver but in motion relative to someone standing outside.



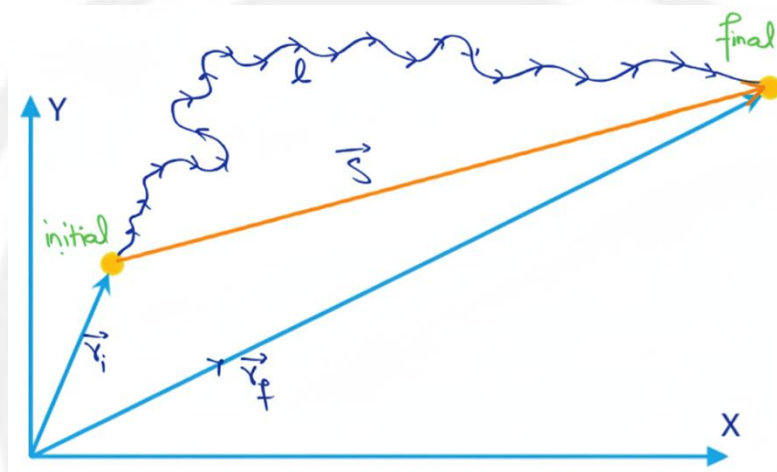
Position of B w.r.t. A ($r_{B/A}$)

- The position of object B with respect to object A is given by the vector difference: $r_{B/A} = r_B - r_A$.
- This relative position vector does not depend on the specific choice of the observer's origin.



Displacement (S) and Distance travelled (l)

- Displacement (S) is a vector quantity, representing the straight-line change in position from the initial to the final point ($S = r_{\text{final}} - r_{\text{initial}}$).
- Distance travelled (l) is a scalar quantity, representing the total length of the path covered by an object.
- The SI unit for both displacement and distance is the meter.



Important points

- Displacement is a vector quantity, while distance travelled is a scalar quantity.
- The magnitude of displacement is always less than or equal to the distance travelled ($S \leq l$).
- Both displacement and distance travelled do not depend on the choice of origin.
- Distance travelled cannot be negative and cannot decrease with time.

